

BAAZ JKBOSE



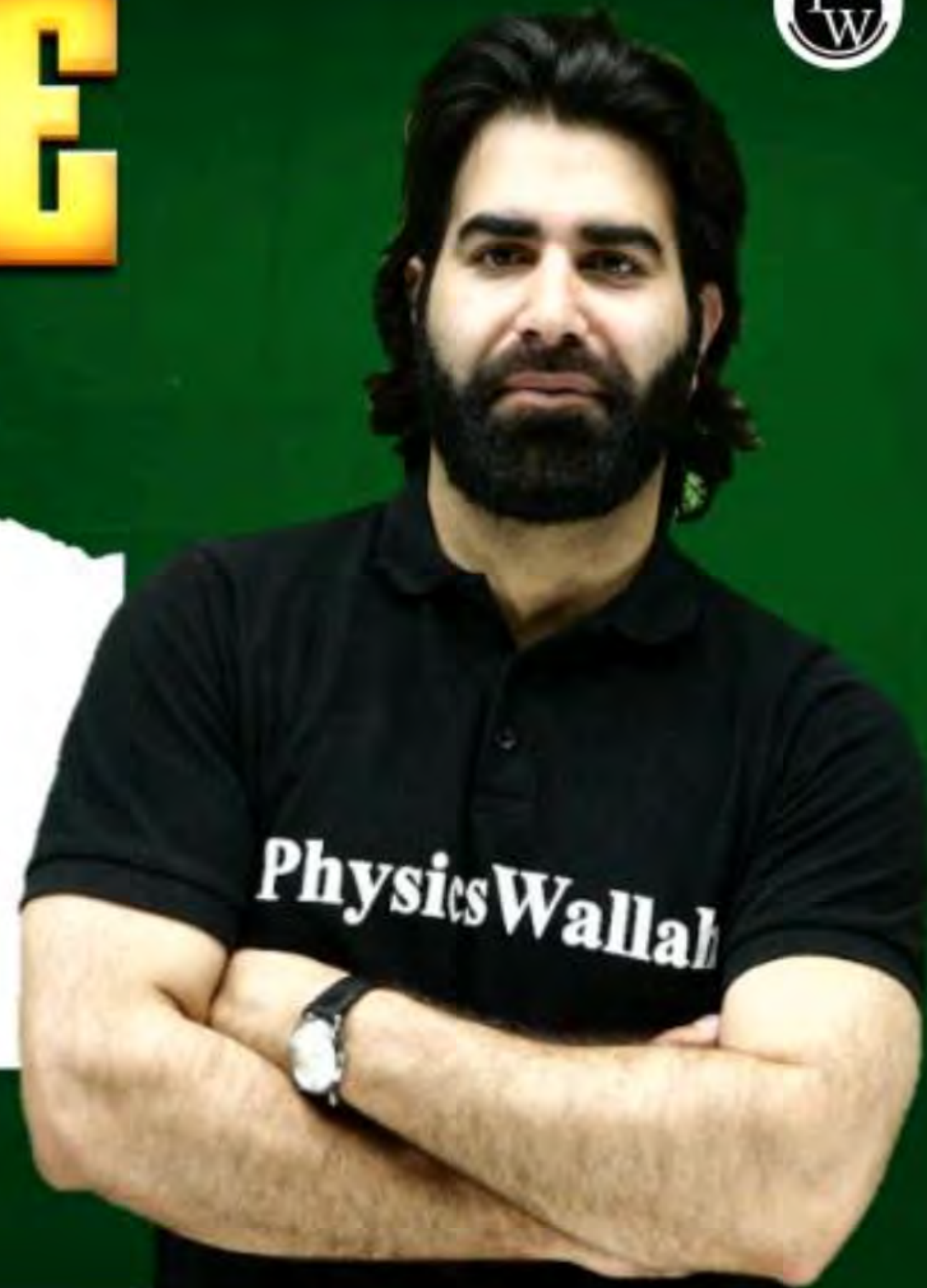
2026 CLASS-11TH

Excretory Products and
Their Elimination

LECTURE-05

Zoology

Dr Adil Altaf sir



Topics To Be Covered

30km Trek

- 1 composition of Urine, Hemodialysis, Disorders.

Revision

→ Mechanism of Conc. of Urine

→ Osmolarity of Cortex 300

→ " " L.O.H 1200
(lower point)

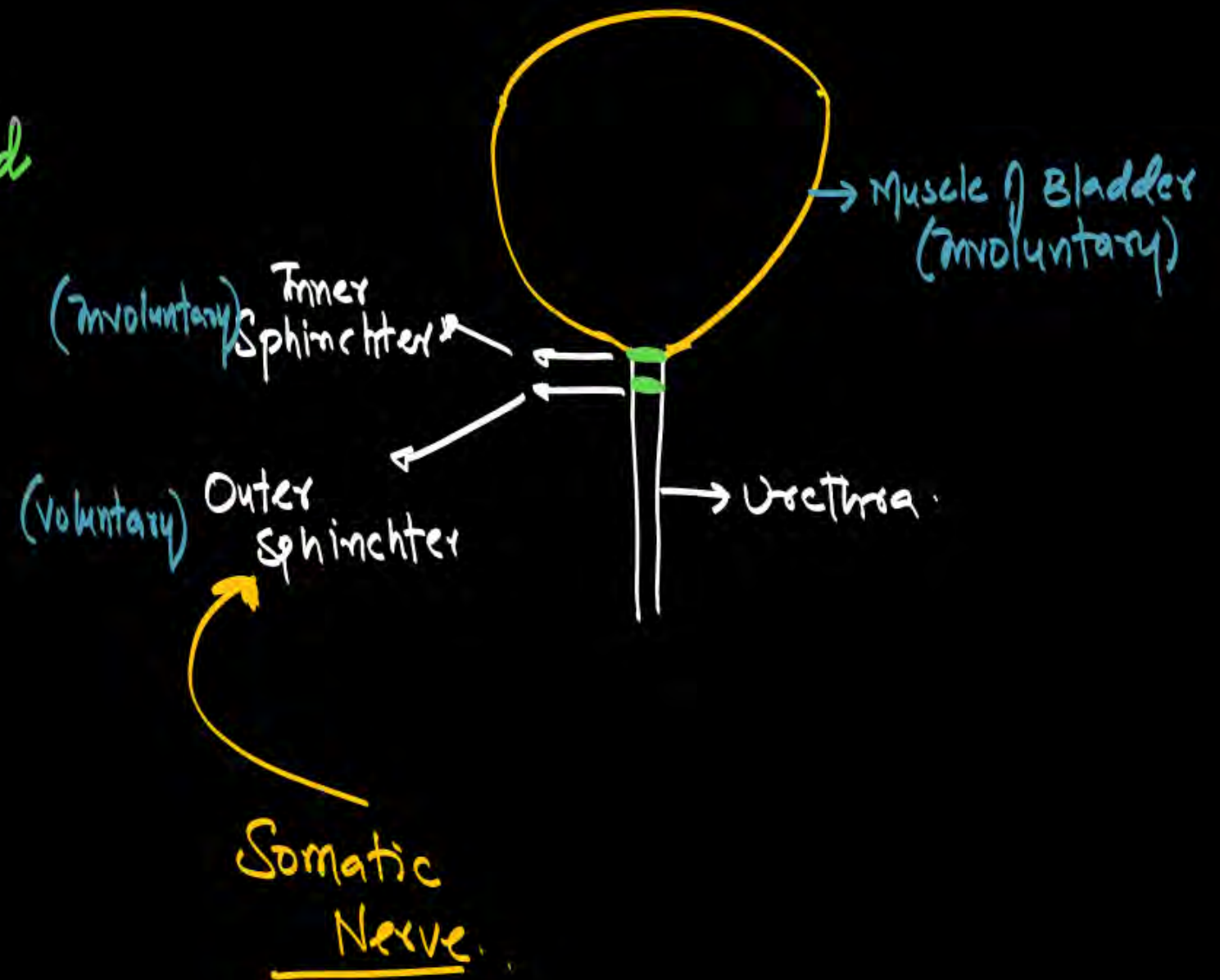
→ ADH Antidiuretic Hormone

• Diabetes insipidus

Cause (low/def of ADH)

Micturition

- ↳ It is a process of Emptying of Urinary Bladder
- ↳ It is a Reflex, but can be controlled upto an Extend
- ↳ It is under Autonomic Nervous System (Involuntary)
- ↳ Parasympathetic Division - Empties U. Bladder
 - Contraction of U. Bladder
 - Relaxation of Sphincters
- ↳ Sympathetic Nervous System - Filling of U. Bladder
 - Relaxation of U. Bladder
 - Contraction of Sphincter



Composition of Urine

↳ Concentrated urine is 95% Water & 5% Solids.

↳ 5% Solids include - 3% N₂ waste (Mainly Urea, Uric Acid, Ammonia, Creatinine)

- 2% Salts (NaCl, PO₄³⁻, Ca²⁺ etc)

↳ Urine - Quantity = (1-1.5 L / Day)
Average.

↳ Urine, Colour - Slightly Yellowish.
- Due to pigment = Urochrome.

↳ Urine, pH = 6. (Slightly Acidic).

↳ Urine, Osmolarity = 1200m Osm/L

Muscles
↓
creatine
↓

Abnormal contents of Urine

- ① Pyuria - Pus cells (Dead NBCs) in Urine (Infection)
- ② Hematuria - Blood/RBC in Urine - (Kidney Infection, Stones, Cancers)
- ③ Proteinuria - Protein in Urine (Infection, Damage to glomerulus)
- ④ Albuminuria (Albumin in Urine)

⑤ Glycosuria - Glucose in Urine (Diabetes mellitus)

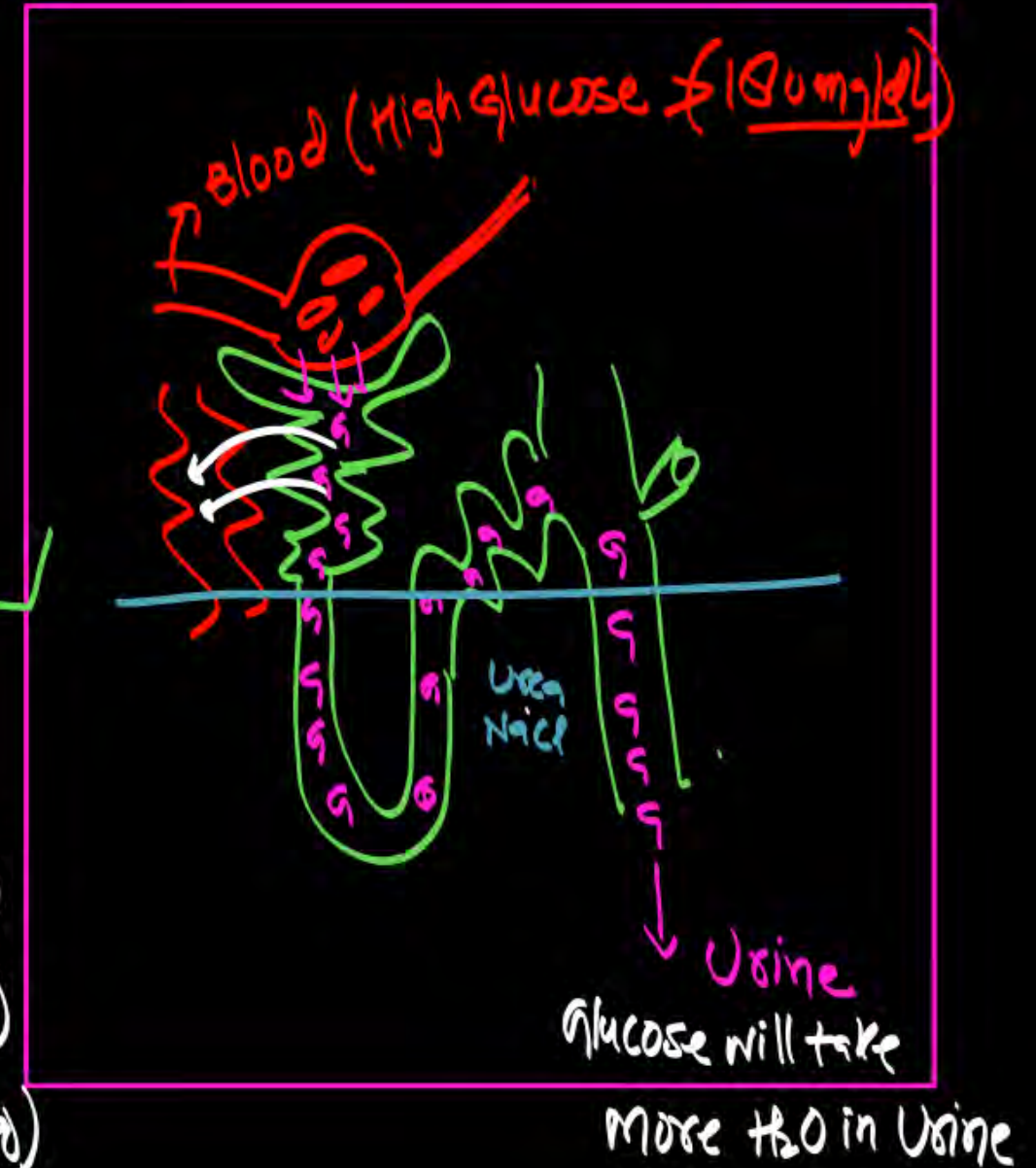
⑥ Ketonuria

↳ Ketone Bodies in Urine

(Diabetes, Starvation)

Why (H/W)

- Symptoms - High glucose level
- polyuria (Large urine production)
 - polydipsia (↑ Thirst)
 - polyphagia (↑ Hunger)



Q1 → What happens at the time of micturition?
(emptying)

23:47



- X (a) U. Bladder Relaxes
- ✓ (b) U. Bladder Contracts
- X (c) Inner sphincter Relaxes only.
- X (d) Both sphincter contract

70

Disorders

→ Kidney Failure - Inability of kidneys to remove sufficient Amount of wastes.
- Leads to Uremia [$\uparrow$ urea in Blood].

- Treatment - Hemodialysis (Artificial kidney). ✓

- Kidney Transplant

↳ Matching of Antigens.

↳ Donor - Close Relatives.

↳ On Immunosuppressants (Medicine)

↳ To suppress Immune System so that Transplanted Organ is not Rejected.

Hemodialysis

↓
Blood

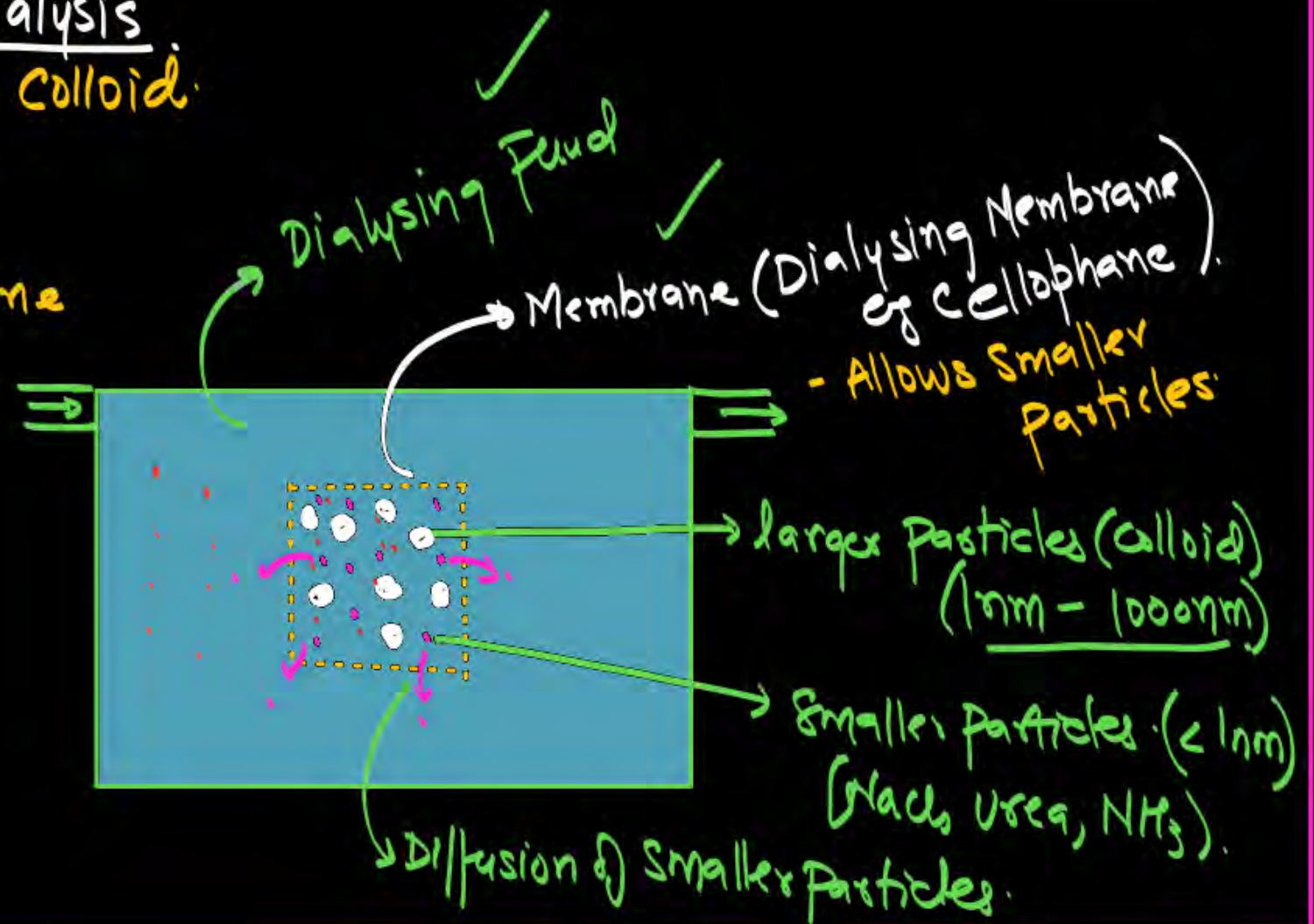
→ Means Dialysis of Blood.

- Used in Kidney Failure Patients
- Also Known as Artificial Kidney

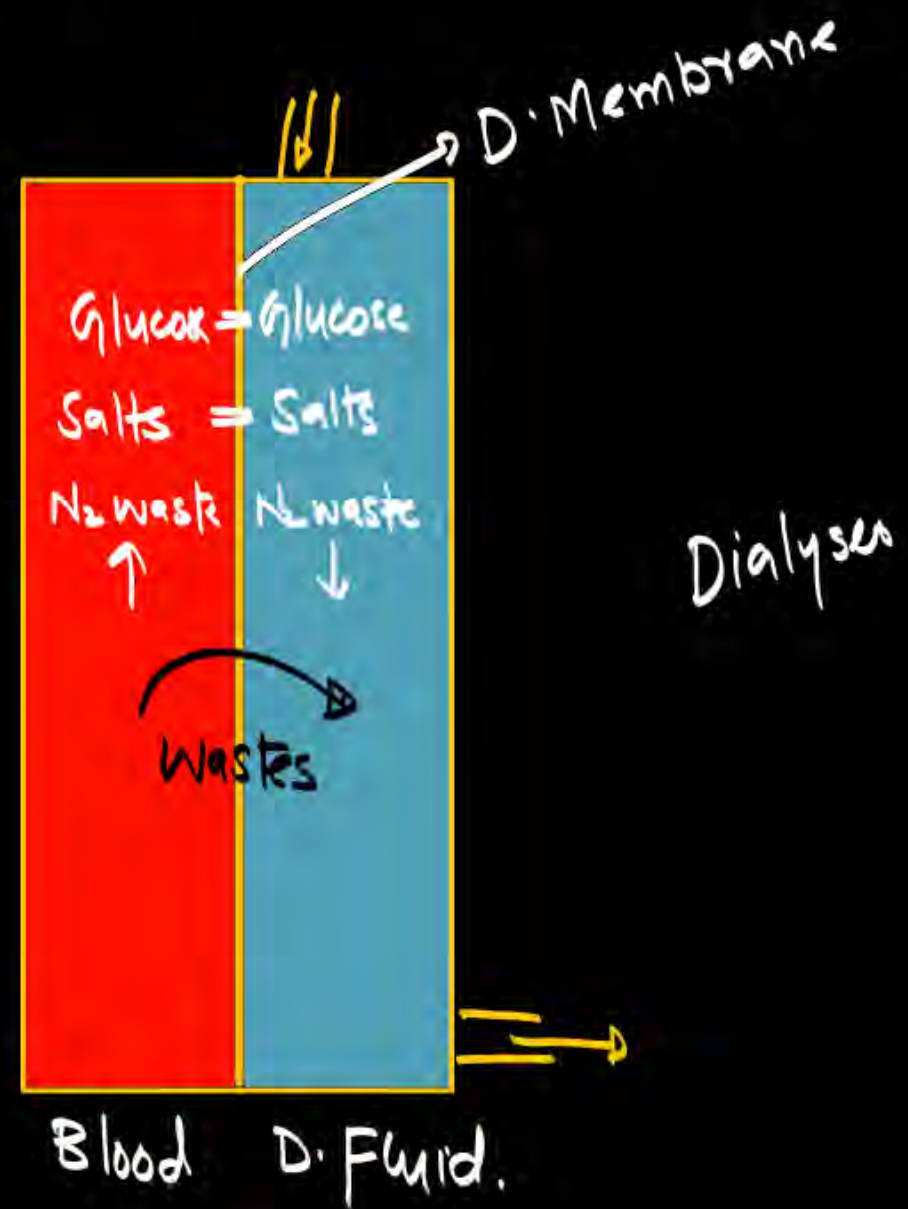
Blood is Colloid

Dialysis

- Dialysis is a Method of Purification of Colloid.
- In Dialysis, the Smaller Particles Move out of Colloid through D. Membrane into D. Fluid By Diffusion.



In Hemodialysis, Blood is given into a Dialyser, where the Dialysing Fluid has same concentration as plasma except H_2O wastes

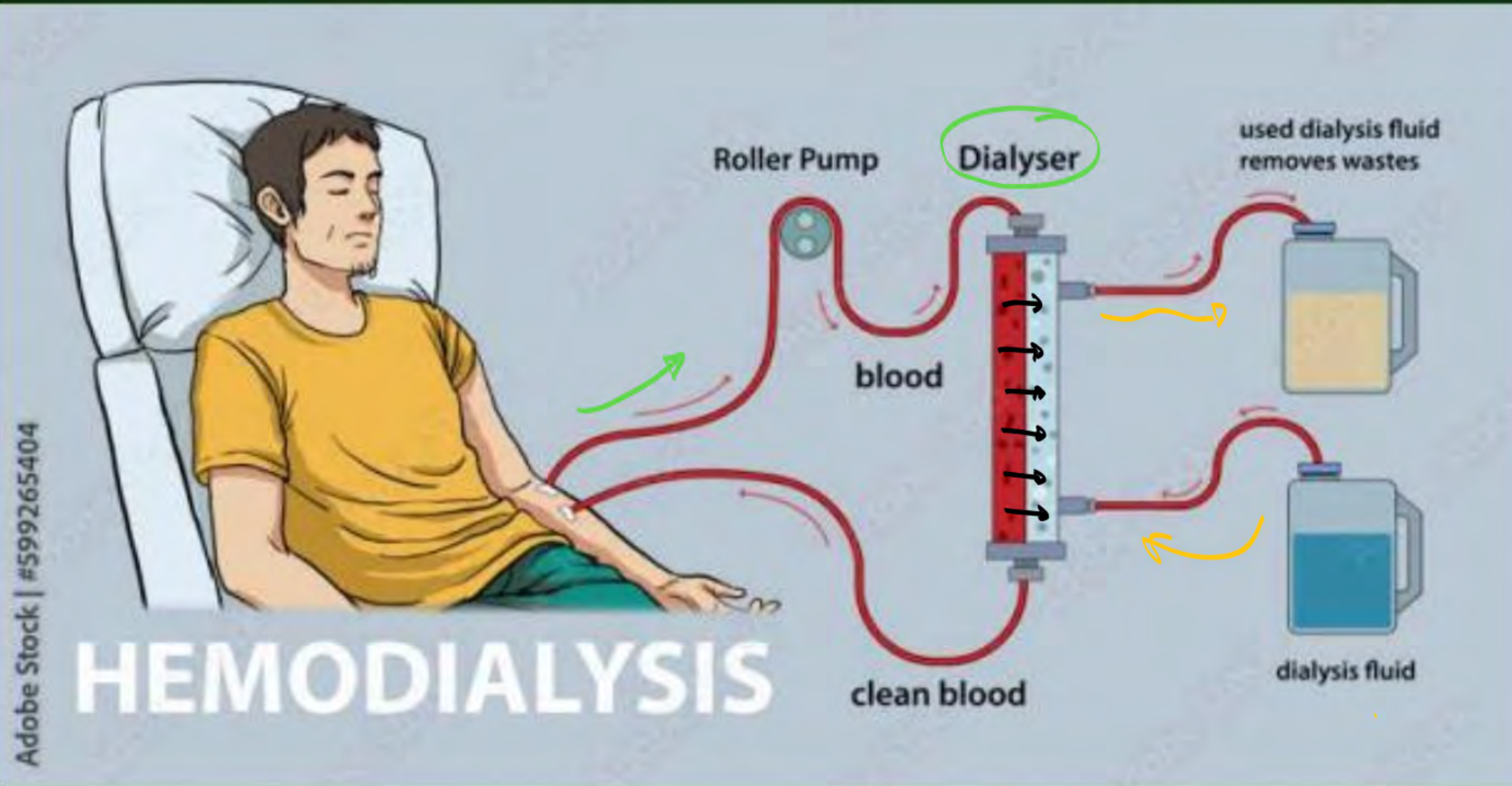


Steps in Hemodialysis.

- ① Blood is taken from artery (Radial Artery)
 - ② Blood is cooled to 0°C .
 - ③ Anticoagulant (Heparin is Added)
 - ④ Given into Dialyser
 - ⑤ Antiheparin (Coagulant is Added) is Added.
 - ⑥ Warm to Body temp.
 - ⑦ Blood given Back to Body Through Radial vein.
- (Neutralise)

(H/W) ⁰Renal Calculi (kidney Stone)

① Glomerulonephritis



Thank
You